Structural reductions for Diophantine consistency equations

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Abstract

This report gives a norm-saturation reduction for a fixed ten-integer Diophantine equation, lowering its exact degree from 1 576 864 to 1 245 324 with 480 arithmetic operations after a certified packing improvement. A separate coefficient-height certificate replaces an entire bounded common-multiple block in the commutative proof source; its degree-oriented local relation has 23 natural witnesses and degree 14 instead of 50. The latter is a proved source-level improvement, not a materialized global consistency polynomial. Completed companions use 37 integers at degree 4, 30 at degree 6, 26 at degree 8, 25 at degree 10, 24 at degree 12, and 23 at degree 16. Positive-domain reductions give 11 variables at degree 277 with 111 operations and 12 at degree 78 with 101 operations. Actual fixed coefficients, literal equations, restoration proofs, independently written audits, and a clean replay archive accompany the report. These are ordinary mathematical reductions over explicitly retained inherited calibrations, not new end-to-end proof-assistant certificates or global minimality claims.

1 Results and exact meaning of the consistency claim

Let $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ be the fixed recursive syntactic refutation predicate used by the retained source calculus, and set

$$\mathrm{Con}(\mathrm{ZFC}) \equiv \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

For every completed polynomial P in this report, the mathematical reduction establishes, in reasoning formalizable in ZFC,

$$\mathrm{ZFC} \vdash \left[\neg\exists \mathbf{x} \in D^n\ P(\mathbf{x}) = 0\right] \iff \mathrm{Con}(\mathrm{ZFC}), \tag{1}$$

where $D = \mathbb{Z}$ or $D = \mathbb{Z}_{>0}$ is stated explicitly. This does not prove Con(ZFC), decide solvability, or assume arithmetic soundness of ZFC. It is a reduction to the same syntactic predicate, not a claim that ZFC proves its own consistency. The inherited source/compiler and Pell calibrations are essential mathematical dependencies. Their formal integration remains distinct from the new algebraic arguments.

Domain and construction	Variables	Exact degree	Operations	Prior comparator
Integer norm saturation	10	1 245 324	480	1 576 864/481
Positive three-root descent	11	277	111	325/113
Positive adaptive saturation	12	78	101	82/102
Integer compatible guards	37	4	—	38/4
Integer centered projection	30	6	—	32/6
Integer exact projection	26	8	—	27/8
Integer transport pivot	25	10	—	separate tradeoff
Integer joint positivity	24	12	—	separate tradeoff
Integer joint positivity and pivot	23	16	—	26/16

For the first three rows the comparator lists degree/operations at the same variable count. For the integer low-degree rows it lists variables/degree. The two 37-variable quartics are distinct equivalent presentations. Low-degree operation schedules have not been optimized or assigned comparative operation claims.

An operation means one binary addition, subtraction, or multiplication; scalar multiplication and squaring count, fixed integer leaves and shared reuse do not. These are arithmetic-circuit counts, not bit complexity. Large fixed coefficients cannot be hidden when comparing overall simplicity. Here they are retained as exact byte payloads, with their recipes, bit lengths and hashes. A positive variable is not an unrestricted integer variable; conversion between those domains is not free.

What is structurally new in this iteration. The ten-variable reduction removes a redundant factor of a Pell norm inside the Lucas square certificate while proving that full residue-class interpolation survives. The commutative source uses deterministic coefficient-height injectivity instead of numerical pairwise coprimality. The low-degree work proves compatible compositions and additional exact eliminations. In particular, the incoming README already suggested the 37-variable quartic composition and a joint-positivity combination. They are not represented here as previously unexplored ideas. The 30-variable sextic and the transport-pivot companions are further completed reductions. No claim of historical priority for a general mathematical technique is made.

Comparison snapshot. The repository snapshot is `2605669a532342f153b13eb6b4a0af4b54f5fa37`. The eight incoming packages 68–75 were compared as complete archives, not merely their placed subdirectories. Nine-positive-variable equations remain outside the improved rows: the inspected degree/cost tradeoffs include $1\,845\,940/444$ and $1\,379\,730/478$. The present result does not reduce the nine-positive or ten-integer variable frontier.

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2 Norm saturation in the ten integer equation

The ten unrestricted variables are

$$(t, g_0, c_0, k_0, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

All other symbols in this section are fixed constants or polynomial abbreviations, not extra quantified coordinates. The literal division-free arithmetic graph is `lucas_h1868/integer10.json`.

2.1 The fixed source and complete equation

Put $\kappa = 2\,134\,940$ and define

$$b = 6t + 4, \quad g = bg_0, \quad k = b(3k_0 + 2), \quad C = 6c_0 - 1, \quad B_0 = 2^\kappa b^2.$$

Let w_i and q_j be the fixed lists in Appendix B, and D_0, D_1, D_2 the three explicitly listed sparse polynomials there. Define

$$\ell_0 = \sum_{i=1}^{31} B_0^{w_i}, \qquad v_0 = \ell_0 + g,$$

$$J_0 = D_2(B_0)v_0^2 + D_1(B_0)v_0 + D_0(B_0), \qquad T_0 = \sum_{j=1}^{22} B_0^{q_j},$$

$$N_0 = B_0^{117}, \qquad N = B_0^{1986},$$

$$M_0 = N_0 - 1 - (b-1)\ell_0,$$

$$S_c = g + N_0(2J_0 + T_0), \qquad T_c = M_0 + N_0(B_0 - 2)T_0 - 2,$$

$$X = N(S_c + T_c + 1) - S_c - 1, \qquad Y = 2^{34\,159\,017} b B_0^{2109},$$

$$\mathrm{cap} = bB_0^{116}.$$

The source height is 1868, certified by all 22 exact extraction identities. Its actual complete coefficient tuple is the retained source-31 tuple. Then set

$$L = lY, \qquad U = 2LX, \qquad V = 4gwY, \qquad W = bw,$$

$$A = U(V+1), \qquad B_1 = 2X + 1, \qquad S = 2A - 5, \qquad T = 3WC - 2(W^2 - 1),$$

$$P_h = (LX)UV, \qquad K = X + 1 + k(2P_h - 2),$$

$$Q = (P_h^2 - 1)K^2 + 1,$$

$$\Delta = 8\mathrm{cap}^3 gK^2 - g^2\{32(C - KL)^2\mathrm{cap}^3 + g^2K^2\}.$$

The new Lucas part is

$$\xi = zx, \qquad D = (A^2 - 4)C^2 + 4,$$

$$E_0 = C^2\xi, \qquad E = DE_0, \qquad E_2 = E_0E = C^4D\xi^2,$$

$$F = 4(A^2 - 4)E_2 + 1, \qquad G = (A/2 - 1)(F - 1) - 1,$$

$$\Omega = 16E, \qquad H = C + F(B_1 - C + \Omega y),$$

$$I = (G^2 - 1)H^2 + 1.$$

Here $A/2 = (LX)(V+1)$ is an integer polynomial abbreviation. No division operation appears in the literal graph. Finally put

$$\rho = T + Sy, \qquad \Sigma = \xi S^5,$$

$$\Theta = (\rho^3 - \rho S^2 - S^3)(\rho^2 + 23S^2),$$
$$r = bQ, \qquad a_0 = r + (9\Delta - 3)(rz)^2, \qquad a_1 = DFI,$$
$$Z = \Sigma m - \Theta.$$

The completed ordinary polynomial equation is

$$\boxed{\left(Z^2 - \Sigma^2(a_0 + a_1)\right)^2 - 4\Sigma^4 a_0 a_1 = 0.} \tag{2}$$

Every exponent is a fixed nonnegative integer.

2.2 The saturation step and reliability theorem

The preceding construction used $E^2 = C^4 D^2 \xi^2$ in the definitions of F and G. Only those occurrences are replaced by $E_2 = C^4 D \xi^2$. The factor D in E and hence in Ω remains. Computing E_0, E, E_2 needs the same three multiplications as the previous E, E^2 schedule. This lowers degree without adding a variable or operation. Applied alone to unchanged height-1873 source data, it gives the control equation in `lucas/` with degree 1 248 364 and 481 operations. The recommended equation additionally uses the certified height-1868 target layout and an exact shared-power plan, reaching 1 245 324 and 480. All twelve fixed coefficient values are unchanged. The source layout changes target positions, not variables, residuals or coefficient ports; its extraction and carry compatibility proof is in `packing/certified/PROOF.md`.

Write $u_0(A) = 0$, $u_1(A) = 1$, $u_{n+2}(A) = Au_{n+1}(A) - u_n(A)$, with the usual extension to integer indices.

Theorem 2.1 (Saturated Lucas reliability)**.** *Suppose A is even, $|A| > 2$, B is odd with $1 < B < |A|/2 - 1$, C is odd with $|C| > 1$, and $\xi \neq 0$. Use the preceding definitions with B replacing B_1. If DFI is a square, then $C = \pm u_B(A)$. On the residue slice $3 \mid A$, $B \equiv 3 \pmod 4$, $C \equiv -1 \pmod 3$, the positive choice $C = u_B(A)$ is forced.*

Proof. The quantities D, E_2, F are positive, $|G| > 1$, and $I > 0$. Because $D \mid E_2$, one has $F \equiv 1$, $G \equiv -1$, $I \equiv 1 \pmod D$. Moreover

$$2G = (A-2)F - A, \qquad 4I \equiv D \pmod F.$$

Since F is odd, these congruences make D, F, I pairwise coprime. A square product therefore makes each a square. Choose in the proof only $d > 0$ with $d^2 = D$. The Pell/Lucas classification yields indices s, μ, t with

$$C = u_s(A), \qquad 4C^2 d\xi = u_\mu(A), \qquad H = u_t(2G).$$

Odd C makes s odd. Since $|C| > 1$, $|s| \geq 3$, and Lucas growth implies $|C| \geq |A|^2 - 1$. Thus $|4C^2 d\xi| > |u_3(A)|$. Consequently $n = |\mu| > 3$ is even, $F = (v_n(A)/2)^2$, and H, t are odd. The index-separation lemma modulo F gives $s \equiv \pm t \pmod{2n}$. The rank-divisibility lemma, using $C^2 \mid u_n(A)$, gives $C \mid n$. Hence $s \equiv \pm t \pmod{2C}$.

For the modified expressions,

$$\frac{F-1}{\Omega\xi} = \frac{(A^2-4)C^2}{4}, \qquad \frac{2G+2}{\Omega\xi} = \frac{(A+2)(A-2)^2C^2}{4},$$

which are integers. It follows that $F \equiv 1$ and $2G \equiv -2 \pmod\Omega$. Therefore $H \equiv B \pmod\Omega$, while $H = u_t(2G) \equiv u_t(-2) = t \pmod\Omega$, because t is odd. Since $2C \mid \Omega$, we obtain $s \equiv \pm B \pmod{2C}$. The inequalities $|s| \leq |C|$ and $B < |A| < |C|$ exclude a nonzero multiple of $2C$ in $s - B$ or $s + B$. Thus $s = \pm B$. Finally $u_{2j+1}(A) \equiv (-1)^j \pmod{A^2}$ gives $u_B(A) \equiv -1 \pmod 3$ on the stated slice, which selects the sign. □

The classical classification, index-separation, and rank-divisibility interfaces are Sun's Lemmas 2.1, 2.2 and Corollary 2.1 in the modified certificate paper. The bound on C above is derived after classification; it is not assumed for arbitrary inputs.

This is a guard-dependent improvement reusable whenever the surrounding arithmetic bridge proves $|C| > 1$. It is not an unconditional replacement of the standalone modified Lucas certificate. The exceptional-index countercontrol in the verification section makes that boundary explicit.

2.3 Full interpolation after saturation

Theorem 2.2 (All residue classes remain available)**.** *Let $A > 2$ be even, $B > 0$ odd, $C = u_B(A)$, and $d = v_B(A) > 0$, so $D = d^2$. For every positive integer N_* there is a positive ξ divisible by N_* such that every class modulo ξ contains a y for which the modified D, F, I are squares.*

Proof. Choose a positive period n of $M_A = \left(\begin{smallmatrix} A & -1 \\ 1 & 0 \end{smallmatrix}\right)$ modulo $4N_*C^2d$. Then $4N_*C^2d \mid u_n(A)$. Put

$$\xi = \frac{u_n(A)}{4C^2d}, \qquad r_0 = \frac{v_n(A)}{2}.$$

These are positive integers, $N_* \mid \xi$, n is even, r_0 is odd, and $F = r_0^2$. Now $\Omega = 4d\,u_n(A)$. The usual Cayley–Hamilton calculation supplies period $\lambda = 4nr_0$ modulo F. Set $q_* = \lambda/\gcd(\lambda, \Omega)$. We claim $\gcd(q_*, \xi) = 1$. For $p \mid \xi$, $F \equiv 1 \pmod{\xi}$ excludes $p \mid r_0$. The Lucas valuation bound $\nu_p(u_n(A)) \geq \nu_p(n)$, including $p = 2$ for even A, yields

$$\nu_p(\lambda) = \nu_p(4n) \leq \nu_p(4u_n(A)) \leq \nu_p(\Omega).$$

Every prime power of λ supported on ξ is absorbed by Ω.

Given a class $c_* \pmod{\xi}$, choose arbitrarily large positive L_* with $q_* \mid L_*$ and $L_* \equiv c_* \pmod{\xi}$. Let $j_* = B + \Omega L_*$ and $H_0 = u_{j_*}(2G)$. Since $\lambda \mid \Omega L_*$ and $2G \equiv -A \pmod{F}$, $H_0 \equiv C \pmod{F}$. Also $H_0 \equiv B \pmod{\Omega}$. As $F \equiv 1 \pmod{\Omega}$, these congruences make

$$y = \frac{H_0 - C - (B - C)F}{\Omega F}$$

an integer and give $H = H_0$. The two quotient identities in the previous proof hold modulo $\Omega\xi$, giving $H_0 \equiv B + \Omega L_*$ and $C + (B - C)F \equiv B$. After cancellation, $Fy \equiv L_* \pmod{\xi}$; hence $y \equiv c_* \pmod{\xi}$. The Pell identity makes I a square. □

Simply substituting $d\xi$ into the previous certificate would not prove this statement: interpolation modulo the new ξ is the substantive compatibility issue resolved here.

2.4 Soundness for every signed assignment

The unchanged source and normalization identities give, before any signs are known,

$$b \equiv Q \equiv 1 \pmod 3, \quad \gcd(b, Q) = 1, \quad 3 \mid A, K, \quad a_0 \equiv 1,\ a_1 \equiv 0 \pmod 3.$$

Thus $a_0 \neq a_1$. The odd number S is nonzero. The intersective polynomial $\pi(v) = (v^3 - v - 1)(v^2 + 23)$ has no rational root, so $\Theta \neq 0$. Equation (2) therefore forces $\Sigma \neq 0$. The two-square norm identity implies that a_0, a_1 are integer squares and $\Sigma \mid \Theta$: for $q = Z/\Sigma$ the equation gives

$$\left(\frac{q^2 + a_0 - a_1}{2q}\right)^2 = a_0,$$

with a complementary rational square equal to a_1; $q = 0$ is excluded by $a_0 \neq a_1$. Rational squares of integers have integral roots, so q is integral. The monic homogeneous quintic's prime-valuation argument then gives $S \mid T$. In particular $\xi, z, x \neq 0$.

Factor $a_0 = r\{1 + (9\Delta - 3)rz^2\}$. The two factors are coprime. The residue $r \equiv 1 \pmod 3$ excludes a negative signed square; hence r is a positive square. Applying the same argument to $r = bQ$ makes b, Q positive squares. If $\Delta \leq 0$, the other factor would be negative, so $\Delta > 0$. The inherited guard bounds imply $K \neq 0$, $0 < g < 2\mathrm{cap}$. If $l = 0$, then $Q = 1 - K^2 \leq 0$, a contradiction. The source bounds now yield $X > 3b$, $Y > 256$, $|L| \geq Y$ and $|A| \geq 2XY > 2B_1 + 2$.

One new estimate removes all exceptional Lucas coordinates:

$$\Delta > 0,\ g \geq 1 \quad \Longrightarrow \quad 4(C - KL)^2 < K^2,$$

so

$$|C| > |K|(|L| - 1/2) \geq Y - 1/2 > 1.$$

Theorem 2.1 now applies, with $B_1 \equiv 3 \pmod 4$, and gives $C = u_{B_1}(A)$. The recurrence implies $A - 2 \mid C - B_1$. Restore in the proof only $h = (C - B_1)/(A - 2)$ and invoke the retained reverse binomial bridge. It decodes the fixed source into a finite ZFC refutation. The old stronger D factor and its exceptional-index argument have not been silently reused.

2.5 Completeness and exact degree

A refutation gives the same canonical arithmetic witnesses as before, with $A > 2$ even, $C = u_{B_1}(A) \equiv 5 \pmod 6$, $r = bQ > 0$ square, $\Delta > 0$ and $S \mid T$. Set $c_0 = (C + 1)/6$. The integer $\delta = 9\Delta - 3$ has 3-adic valuation one, so δr is positive nonsquare. A positive Pell solution $u^2 - \delta r z^2 = 1$ makes $a_0 = ru^2$ a square. Apply Theorem 2.2 with $N_* = z$, put $x = \xi/z$, and use intersectivity of π to choose the required y class modulo ξ. Then $\Sigma \mid \Theta$ and a_1 is a square. Taking

$$m = \Theta/\Sigma + \sqrt{a_0} + \sqrt{a_1}$$

solves (2). This is an existential equivalence, not a bijection between all old and new witness tuples.

For exact degree let $a = \deg A = 16\,384$; the unchanged source has $\deg X = 7942$, $\deg Y = 4219$, $\deg \mathrm{cap} = 233$. The relevant degrees are

	D	E_2	F	G	Ω	H	I	a_1
deg	$2a + 2$	$2a + 10$	$4a + 10$	$5a + 10$	$2a + 6$	$6a + 17$	$22a + 54$	$28a + 66$

and $\deg \Sigma = 5a + 2$. The guard degree 294 619 and center degree 81 925 are below their competitors. The dominant final term is $\Sigma^4 a_1^2$, yielding

$$\deg P_{10} = 4(5a + 2) + 2(28a + 66) = 76a + 140 = 1\,245\,324.$$

Independent nonzero top-coefficient evaluations modulo three primes certify attainment on the actual fixed-coefficient slice. The final graph has 300 multiplications, 155 additions and 25 subtractions: 480 operations.

3 Compatible sign recovery and low-degree integer equations

The incoming weighted-cone and residue-orientation arguments can be applied simultaneously, but this requires a completeness argument: both are restrictions on canonical witnesses, not identities on all old

solutions. The resulting fixed-degree improvements are

$$(38,4) \longrightarrow (37,4), \qquad (32,6) \longrightarrow (30,6), \qquad (27,8) \longrightarrow (26,8),$$

where each pair means number of unrestricted integer variables and total degree. Further exact projections give $(25,10)$, $(24,12)$ and $(23,16)$. All equations are sums of explicitly listed residual squares. Their literal packets, six actual coefficient payloads, full restoration maps and proofs are in `positivity/`.

3.1 The common quartic source

Use the whole source-34 coefficient tuple, including its already shifted mask M_F, and fixed weights $A = 2^{1113}$, $B = 2^{666104}$. Write

$$q = \beta J + 1, \quad C = q - F - \alpha, \quad T = AFBC - (q - AF - BC)^2, \quad \mathrm{Pos}(p,q,r) = 1 + p^2 + q^2 + r(r+1).$$

The source has the two sign rows $T = \mathrm{Pos}(\mathbf{t})$ and $\eta\zeta = \mathrm{Pos}(\mathbf{e})$, each using three private integer coordinates. If $T > 0$, then AF and BC have one nonzero sign and

$$|q - AF - BC| < \sqrt{AFBC} \le (|AF| + |BC|)/2.$$

Since $A, B \ge 2$, this forces q, F, C, α to share one strict sign. Thus the older positive products $F\alpha$ and FC are recoverable. Conversely the retained canonical support-and-carry proof gives $q/2 \le AF < 4q/5$ and $q/2 \le BC < 3q/5$, uniformly under the permitted rank-tuning changes. Hence every canonical accepting witness has $T > 0$.

Now delete the square guard $X = 2u^2$ and its private unknown u. The compact variant substitutes $X = 3X_3 - 1$. The projection-friendly variant instead substitutes

$$J = 3J_3 + 1, \qquad w_{\mathrm{scale}} = 3w_3 + 1.$$

Here $\beta = 2^{1064921} - 1 \equiv 1 \pmod 3$, so $q \equiv 2$, $W = 4(3w_3 + 1) \equiv 1$, and $X = Wq \equiv 2 \pmod 3$. The inherited signed Pell argument recovers $X = \epsilon 2^R$ with R odd before imposing a sign on X. Its residue therefore selects $\epsilon = 1$, restoring $u = 2^{(R-1)/2}$. The determinant guard supplies exactly the older common-sign hypotheses needed by that argument. For the direct variant $X \ne 0$ and $X = 4qw_{\mathrm{scale}}$ give $|4w_{\mathrm{scale}}| \ge 4$, as required by the same signed proof.

Completeness of the combined restrictions is important. The canonical family has $q = 2^t$ with t odd and $w_{\mathrm{scale}} = 2^{R-t-2}$ with nonnegative even exponent. Thus $J = w_{\mathrm{scale}} = 1 \pmod 3$. The cone estimate is uniform over this same rank-tuned family, rather than over an incompatible family. Consequently both 37-variable quartics have the required consistency calibration. Their τ^4 coefficient is one.

3.2 An ideal-preserving sextic projection

For the affine quartic let $k = \eta + \zeta$ and $H = 4a + 3$. Center two roots by

$$\tau_{\mathrm{old}} = \tau + E(c - \eta), \qquad D_{\mathrm{old}} = D + ac,$$

and eliminate seven monic registers:

$$P = k + E(c - \eta), \quad X = Wq, \quad N = q(L+1), \quad C_2 = c^2, \quad \Delta = a^2 + 4a + 3, \quad V = fo - c, \quad B_{\mathrm{aux}} = I^2.$$

Keep $F_2 = f^2$. Replace the restored row $V^2 + g - y^2$ by

$$F_2 o^2 - 2foc + c^2 + g - y^2.$$

Their difference is $o^2(F_2 - f^2)$, a multiple of a retained equation. The two root translations give exact cancellations

$$\begin{aligned}(\tau + E(c-\eta))^2 - (k + E(c-\eta))E(c-\eta) - 1 &= \tau^2 + (2\tau - k)E(c-\eta) - 1,\\ (D + ac)^2 - (a^2 + 4a + 3)c^2 - 1 &= D^2 + 2acD - (4a+3)c^2 - 1.\end{aligned}$$

All 17 resulting residuals are at most cubic, giving a sextic in $37 - 7 = 30$ unrestricted integers. The coefficient of $I^4 g^2$ is one. The polynomial restorations, root translations and retained F_2 row prove an integer zero-set bijection, not just an equisatisfiable heuristic.

3.3 Transport pivots and a joint-positive-value row

The ordinary eleven-register projection of the affine quartic gives 26 unknowns, 13 quartic residuals, and an octic. Put $k_0 = 1 - V + jc + hE$; its two sign quantities are T and $Z = \eta(k_0 - \eta)$. The coefficient $[a^4 f^4]$ of the octic is one.

Replacing α by $C = q - F - \alpha$ makes the transport row monic:

$$(K_0 + W)C - F - z(q-1) = 0.$$

Eliminate $F = (K_0 + W)C - z(q-1)$ exactly. The packing polynomial becomes

$$(q - F)(q^3 - q) - C(q^2 - 1) + (M_C + qM_F)(3J_3 + 1),$$

of degree at most five. The result has 25 unknowns and degree ten; its coefficient at $w_3^2 C^2 J_3^6$ is $104976\beta^6 > 0$.

For arbitrary integers x, y the nearest-square lemma is

$$x, y > 0 \quad \Longleftrightarrow \quad \exists t \in \mathbb{Z} : \quad G(x,y,t) := xy - (t^2 - x - y + 1)^2 > 0.$$

For soundness, $G > 0$ implies $xy > 0$, while two negative arguments make the squared term larger than xy. For completeness set $n = x + y - 1$; then $1 \le n \le xy$, and a nearest square has squared error less than n, unless the error is zero. This proves $G > 0$.

Replace the two independent sign triples in the 26-variable octic by

$$G(T, Z, t) = \mathrm{Pos}(p, q_0, r).$$

There are now 24 unknowns and 12 residuals. Since T is quadratic and Z cubic, this equation has degree twelve, certified by the coefficient $[\eta^4 j^4 c^4] = 1$. Applying the same transport pivot gives 23 unknowns and 11 residuals of total squared degree sixteen. Its coefficient $[w_3^8 C^8] = 429981696A^8$ is positive on the actual fixed tuple. These two sign substitutions preserve existence by explicit witness selection and restoration; they are not bijections on private witnesses.

The component checks exact algebra, liveness, degree certificates, actual coefficient payloads and independently coded compact formulas. Those finite checks supplement the unbounded proofs; they do not establish native compiler semantics or an end-to-end proof-assistant theorem.

4 Rational descent and the positive domain

The two equations in this section have strictly positive integer variables. Both use the complete compact44 coefficient tuple, with $\beta = 2^{590069} - 1$, K_0 its K port and M_F its already shifted $M_{F,\mathrm{source}}$ port. The whole tuple is fixed by `positive/native/ACTIVE_BINDING.json`; none of the ports is mixed with the larger tuple used in the low-degree integer equations.

4.1 A denominator theorem for adjacent Pell coefficients

Lemma 4.1. *If $Q > 1$ and V are integers, $y \in \mathbb{Q}$, and $QV^2 - (Q-1)y^2 = 1$, then $y \in \mathbb{Z}$.*

Proof. Take absolute values, so $V, y > 0$; zero is impossible. Write $y = a/m$ in lowest terms, $m > 0$. The equation implies $m^2 \mid Q - 1$. If $V = 1$, then $y = 1$. Otherwise $y > V$ and $QV > (Q-1)y$. Define

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

The first quantity is integral, the norm remains one, and the strict inequalities give $-V < V' < V$. The norm excludes $V' = 0$. The reduced denominator of y' is still m, since $2Q - 1 \equiv 1 \pmod{m}$. Strong induction on $|V|$ reaches $V = 1$ without changing the denominator, so $m = 1$. □

This special descent replaces an auxiliary prime-support requirement. It is not a claim that general rational Pell coordinates are integral.

4.2 Eleven positive variables without prime support

Use the eleven variables

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma > 0.$$

The complete register definition is

$$\begin{gathered}
J = 3j + 1, \qquad q = \beta J + 1, \qquad C = q - F - \alpha, \\
E = XY, \qquad k = \eta + \zeta, \qquad c = kY + \eta, \\
a = E + Y, \qquad H = 4a + 3, \qquad \Delta = a^2 + H, \\
\ell = EkY, \qquad M = q^3(q-1), \qquad B_t = (qK_0 + X)C - qF, \\
R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\
\widehat{R} = q^3(q + F + \alpha), \qquad D_\tau = \ell(\ell + k) + 1, \\
I = ic^2, \qquad D_f = 1 + \Delta I^2, \qquad Q = \Delta^2 I^2, \\
\delta = Q - 1, \qquad V = D_f(cT - R) - c, \qquad D_{\mathrm{num}} = QV^2 - 1, \\
\overline{V} = D_f(cT + \widehat{R}), \qquad L = 3\overline{V}, \qquad W = 1 + (q-1)L, \\
\overline{Z} = M(\ell + k + n + L) + (q-1)Y - q^3 W B_t.
\end{gathered}$$

The adaptive main carrier and three-root norm are

$$\begin{gathered}
r_0 = k - R - 1, \qquad S_0 = 4qY, \qquad U = 4q^2 Y, \\
D_* = U(X + ac) + H(U\gamma + S_0 X - r_0), \\
A_1 = D_*^2 - U^2(1 + \Delta c^2),
\end{gathered}$$

$$A_N = M^2 D_\tau, \qquad B_N = M^2 D_f, \qquad C_N = M^2 D_{\mathrm{num}},$$
$$N_c = \delta^2\{(\overline{Z}^2 - A_N - B_N)^2 - 4A_N B_N\}$$
$$-2\delta C_N(\overline{Z}^2 + A_N + B_N) + C_N^2,$$
$$N_8 = N_c^2 - 64\delta^3 A_N B_N C_N \overline{Z}^2.$$

The literal ordinary equation is

$$P_{11} = (1 + N_8)(1 + A_1) - 1 = 0. \tag{3}$$

Its graph is `positive/p11_degree277.json`. The only change to the retained three-root source is $I = ic^2$ in place of $iq(q-1)c^2$.

Soundness. The retained unconditional estimates need only $I \geq c^2$, so they remain valid. In particular

$$1 < D_\tau < D_f < Q, \quad |V| > 1, \quad D_y = \frac{QV^2 - 1}{Q - 1} > V^2 > 1.$$

The three positive rational radicands D_τ, D_f, D_y are distinct, and no signed sum of their roots is zero. To recall why, if $cT - R = 0$ then $V = -c$ and $\sqrt{D_f} > \sqrt{D_\tau} + \sqrt{D_y}$. Otherwise $|V| \geq D_f - c$ gives the reverse strict dominance $\sqrt{D_y} > \sqrt{D_f} + \sqrt{D_\tau}$. The fixed-mask bound $|R| < \widehat{R}$ gives $L > \sqrt{D_f} + \sqrt{D_y}$ and $L > k$.

The octic norm identity makes $N_8 = 0$ imply a rational signed sum $\overline{Z}/M = \tau_* + f_* + y_*$ of those roots. For completeness, rational extraction is elementary: if signed roots x_1, x_2, x_3 with distinct squares a, b, c have nonzero rational sum t, put

$$u = (t^2 - a - b - c)/2, \qquad v = (u^2 - ab - ac - bc)/(2t).$$

Then $u = x_1x_2 + x_1x_3 + x_2x_3$, $v = x_1x_2x_3$, and

$$x_1 = \frac{a(a+u)}{at + v}, \qquad at + v = x_1(x_1 + x_2)(x_1 + x_3) \neq 0.$$

All three roots are rational. The first two are integral because their squares are integers. Apply Lemma 4.1 to the third, with the unconditionally integral Q, V. It too is integral. This is exactly the step that removes $q(q-1)$ from I; no carrier divisibility or later native calibration has been assumed.

Restore positive witnesses $i_{\mathrm{old}} = i$, $f_{\mathrm{old}} = |f_*|$, $u = |y_*| - V$, and $r = n + L - f_* - y_* > 0$. The signed roots, not their absolute values, must be used in r. The first carrier becomes $M\tau_*$, and the four residuals of the retained weighted transport interface vanish. That interface restores the positive native source and hence a finite refutation. Finally N_8 and A_1 are squares modulo 4 (with $4 \mid U$). Their unit-product finalizer cannot use a factor -1, which would require a residual $-2 \pmod 4$. Thus (3) forces $N_8 = A_1 = 0$ without assuming the residuals nonnegative.

Completeness. Choose the canonical outer source. A positive Pell pair can be chosen with I divisible by c^2, because the determinant-one Pell matrix has finite order modulo c^2. The retained auxiliary reconstruction needs only $c \mid I$. It supplies positive T, u; choose L, W afterwards, then the transport witness r. Its bound $r > L$ ensures $n = r - L + f + y > 0$. This gives $N_8 = A_1 = 0$. Alternatively the more divisible old pair works with new $i = q(q-1)i_{\mathrm{old}}$. Either argument proves existence, not equality of the old and new coordinate assignments.

Exact degree and cost. The successive exact degrees of $I, D_f, Q, V, L, W, \overline{Z}, N_c, N_8, A_1$ are $5, 14, 18, 18, 18, 19, 24, 132, 264, 13$. Put $q_0 = 3\beta j$, $k_0 = \eta + \zeta$. The top forms are

$$Q_0 = i^2 k_0^4 X^4 Y^8,$$
$$\overline{Z}_0 = -3i^2 q_0^7 k_0^4 X^2 Y^6 (q_0 + F + \alpha) B_{t,0},$$
$$[A_1]_{13} = 32 q_0^2 k_0 X^2 Y^4 (4 q_0^2 Y \gamma + q_0^3 (q_0 - F)),$$

where $B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F$. Only $Q_0^4 \overline{Z}_0^8 [A_1]_{13}$ reaches degree 277. The coefficient of $j^{60} i^{24} \eta^{49} X^{42} Y^{85} \alpha^{16} \gamma$ is $128 \cdot 3^8 (3\beta)^{60} > 0$, independently of the other three ports. The two deleted multiplications reduce the cost from 113 to 111.

4.3 Twelve positive variables with an adaptive saturated carrier

Use

$$j, F, \alpha, t, T, X, Y, r, \eta, \zeta, n, \gamma > 0.$$

Define $J = 6j + 1$, $f = t + 1$, and the same $q, C, E, k, c, a, H, \Delta, \ell, R, B_t$ as above with this new J. Now put

$$v = k - R - 1, \qquad U = q^2 Y,$$
$$D_* = U(X + ac) + H(U\gamma + qXY - v),$$
$$W = (q-1)k + 1, \qquad M = q^3 (q-1),$$
$$Z = M(\ell + k + r) + (q-1)Y - q^3 W B_t,$$
$$A_0 = Z^2 - M^2\{\ell(\ell + k) + 1\}, \qquad A_1 = D_*^2 - U^2(1 + \Delta c^2),$$
$$\rho = f^2 - 1, \qquad d = \Delta c^4, \qquad Q = \Delta \rho,$$
$$V = f^2(cT - R) - c,$$
$$h(s) = (1 - Q)s^2 + 2(1 - Q)Vs + V^2 - 1,$$
$$a_N = h(n), \qquad b_N = 2(1 - Q)(n + V),$$
$$A_N = d a_N + (1 - Q)\rho, \qquad N_* = A_N^2 - d b_N^2 \rho.$$

The complete equation is

$$P_{12} = (1 + A_0)(1 + A_1)(1 + N_*) - 1 = 0. \tag{4}$$

Its graph is `positive/p12_degree78.json`. Compared with source69, $U = q^5$ becomes $q^2 Y$, and $q^4 X$ becomes qXY in D_*. No new unknown is introduced.

Soundness in the required logical order. The parity argument is unchanged: q is even, so M, U are even, and the three residuals are squares modulo 4. The product finalizer therefore forces all three to vanish. The first carrier, without the strong or auxiliary equation, gives $q^3 \mid Y$, $q - 1 \mid B_t$, $C > 0$, and $3q < R < q^4 - 2$. Thus $U \geq q^5$ and the original positivity bound for D_* survives. The first and main Pell classifications give the first index at least two and $c > 8XY^3 > 2R$. Hence $V > 1$ before $q \mid X$ has been recovered.

If $N_* = 0$, the proof-only rational number $i = A_N/(d b_N)$ satisfies $di^2 = \rho$ and $h(n - i) = 0$. Set $y = V + n - i$; then $QV^2 - (Q-1)y^2 = 1$. Lemma 4.1 makes y, then i, integral. Taking appropriate absolute values restores positive strong and auxiliary witnesses. The source73 adaptive transport theorem applies because it uses $q^3 \mid Y$ to show $U \mid 4XY^2$, not a circular assumption $q \mid X$. It restores the remaining positive source and refutation.

Completeness. The canonical J is 1 (mod 6), so $j = (J-1)/6 > 0$. The source73 lift chooses $\gamma = \gamma_{\mathrm{old}} - w + (k-R-1)/U > 0$, since $(k-R-1)/U \geq 4XY/q^2 > w$. Let $t = f - 1 > 0$ and $n = i + u > 0$. The identity

$$N_*|_{\rho=di^2} = d^2 h(n-i)h(n+i)$$

gives $N_* = 0$, while the carrier residuals vanish. The dependence on the retained native calibration is the same as in the source69/source73 composition, not a new formal certificate.

Exact degree and cost. With $q_0 = 6\beta j$, $k_0 = \eta + \zeta$, $R_0 = q_0^3(q_0 - F)$ and $B_0 = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F$,

$$[A_0]_{15} = -2q_0^8 k_0^2 X Y^2 B_0,$$
$$[A_1]_{13} = 8q_0^2 k_0 X^2 Y^4 (q_0^2 Y\gamma + R_0),$$
$$[N_*]_{50} = 4k_0^8 X^8 Y^{16} t^8 R_0^2 n^2.$$

Their product is the unique degree-78 contribution. Its coefficient at $j^{18}\eta^{11}X^{12}Y^{23}t^8 n^2 \gamma\alpha F^2$ is $64(6\beta)^{18} > 0$, independent of the other ports. The literal evaluator has 56 multiplications, 26 additions and 19 subtractions: 101 live operations.

5 A coefficient height register removes the common multiple block

The previous commutative source assigned label j the factor $1 + jM$, then required M to be divisible by every positive integer up to a label bound. That made the evaluated factors pairwise coprime, but introduced a second bounded-universal certificate. The following alternative dispenses with pairwise coprimality entirely. Its arithmetic ingredient is an elementary bounded-height root argument; the proposed improvement is its application to this proof source, not a claim of historical priority for that argument.

Polynomial identity from one integer evaluation. For a finite list of padded rules (p_i, q_i, c_i), $i < N$, put

$$A(X) = \prod_{i<N} (1 + c_i X), \qquad B(X) = (1+X) \prod_{i<N} (1 + p_i X)(1 + q_i X),$$
$$R = 2 \prod_{i<N} \big((1+p_i)(1+q_i) + c_i\big).$$

All labels are nonnegative. If $M > R$, then

$$A(M) = B(M) \quad \Longleftrightarrow \quad A = B \text{ in } \mathbb{Z}[X].$$

Indeed, each factor in the definition of $R/2$ is at least both $1 + c_i$ and $(1+p_i)(1+q_i)$. Thus $A(1) \leq R/2$ and $B(1) \leq R$. Their coefficients are nonnegative, so every coefficient of $A - B$ has absolute value at most $R < M$. If a_k were its first nonzero coefficient, $(A-B)(M) = 0$, divided by M^k and reduced modulo M, would give $M \mid a_k$, a contradiction.

Distinct positive labels j give distinct rational roots $-1/j$ of the linear factors. The polynomial identity therefore says exactly

$$\#\{i : c_i = j\} = \mathbf{1}_{j=1} + \#\{i : p_i = j\} + \#\{i : q_i = j\} \qquad (j > 0).$$

Zero labels contribute the unit factor. Consequently the same single-conclusion Horn-flow theorem recovers derivability of target 1. For the calibrated numerical C222 calculus this is equivalent in ZFC to a ZFC refutation. There is no separate primality or coprimality hypothesis.

Two registers and no label bounds. Use natural registers S_i, R_i and require

$$S_0 = M + 1, \quad S_N = 1, \quad R_0 = 2, \quad M = R_N + 1,$$

$$\begin{aligned} S_{i+1}(1 + c_i M) &= S_i(1 + p_i M)(1 + q_i M), \\ R_{i+1} &= R_i\big((1 + p_i)(1 + q_i) + c_i\big). \end{aligned}$$

The first recurrence preserves strict positivity from $S_0 > 0$, so multiplying and cancelling yields $A(M) = B(M)$. The second recurrence provides the required coefficient bound exactly. This proves soundness without any chronological semantics for the encoded list.

For completeness, fix a finite proof tree and list its nodes by expanding open obligations. Compute the height product first, set $M = R_N + 1$, and then let S_i be the product of $1 + jM$ over its open obligations. Both recurrences and all endpoints follow. Choosing M after the finite rule list but before the S_i avoids a circular bound on a code already containing M.

The literal local polynomial. Let $H(p, q, c; \mathbf{w})$ be the inherited nonnegative degree-50 Horn polynomial, with seventeen local witnesses. With $M = M_0 + 1$, define

$$\begin{aligned} E &= S'(1 + cM) - S(1 + pM)(1 + qM), \\ F &= R' - R\big((1 + p)(1 + q) + c\big), \\ L &= H + E^2 + F^2. \end{aligned}$$

This uses twenty local natural witnesses, namely $p, q, c, \mathbf{w}$, and five exposed coordinates S, S', R, R', M_0. Its exact degree remains 50: E has degree five and F degree three, so the additional squares cannot cancel the degree-50 homogeneous part of H. The JSON supplies every Horn factor and both new residuals literally.

One bounded block. Write $\beta(x, y, i) = x \bmod (1 + (i + 1)y)$. The five outer coordinates are N, x, z, y, M_0. The endpoint conditions are

$$\beta(x, y, 0) = M_0 + 2, \quad \beta(x, y, N) = 1, \qquad \beta(z, y, 0) = 2, \quad \beta(z, y, N) = M_0.$$

The only bounded-universal block is

$$\forall i < N \; \exists \mathbf{u} \in \mathbb{N}^{20} : \quad L\big(\beta(x, y, i), \beta(x, y, i + 1), \beta(z, y, i), \beta(z, y, i + 1), M_0; \mathbf{u}\big) = 0.$$

Both finite register sequences admit a common y: take y larger than all their entries and divisible by $N!$, and apply the Chinese remainder theorem separately to x and z. This divisibility is used only to prove existence of the beta encoding. It is not an extra input condition and it is not a condition on M.

Compared with the preceding source, the label-bound parameter, three local slack witnesses, and the entire common-multiple bounded block are removed. A second accumulator stream is introduced. The number of outer parameters stays five. The supplied closed Lean constructor uses the current remainder, bounded-forall, and quartic APIs, but has not been elaborated or expanded. These are source-level counts, *not* a claim that the completed consistency equation has twenty variables or that its compiled size beats the existing frontier.

Checks and limitations. The independent finite checks include 24 844 height fixtures, 100 constructed proof-tree paths, twelve simultaneous beta encodings, 800 literal residual comparisons, and twelve exact inherited Horn sum-of-squares identities. At $M = 1$, the rule with premise 1 and conclusion 3 supplies the spurious unchecked accumulator path $2 \to 1$; the new height bound rejects it because $R_N = 10$. This negative control shows why discarding the common-multiple block without replacing its collision protection would be unsound. The global ordinary polynomial remains unmaterialized, and no new proof-assistant certificate is claimed.

A degree-oriented selector variant. There is a further local reduction without increasing the old source's local witness count. The twelve inherited branch factors have degrees $2, 4, 4, 4, 4, 4, 4, 4, 4, 8, 4, 4$. Partition them into six groups of degree 8, the truth group of degree 2, and one unused group assigned value 1. Let $G_0, \dots, G_7$ be these nonnegative products. Introduce three selectors b_0, b_1, b_2 and their multilinear Boolean indicators

$$\chi_i(\mathbf{b}) = \prod_{k=0}^{2} \begin{cases} b_k, & \text{bit } k \text{ of } i \text{ is } 1, \\ 1 - b_k, & \text{otherwise.} \end{cases}$$

Replace the Horn product by

$$\sum_{k=0}^{2} [b_k(b_k - 1)]^2 + \sum_{i=0}^{7} G_i \chi_i(\mathbf{b})^2.$$

All summands are nonnegative. A zero forces Boolean selectors and hence one vanishing selected group; the unused group cannot be selected. Conversely, any vanishing original branch admits a selector assignment. Its exact degree is $8 + 6 = 14$. The leading terms cannot cancel because they are the common monomial $b_0^2 b_1^2 b_2^2$ times a sum of nonzero nonnegative leading forms of products of sums of squares. Adding E^2 and F^2 preserves this degree.

Thus the degree-oriented local source has twenty-three witnesses, matching the old source, but degree fourteen instead of fifty and only one bounded block instead of two. A separate one-selector tradeoff uses twenty-one witnesses at degree twenty. Both have literal JSON and uncompiled constructor files. Their finite selector audits cover 34 768 and 45 556 assignments respectively; all three generated constructors also pass bracket and variable-range checks, which are not Lean elaboration.

6 Evidence and reproducible verification

6.1 What the checks establish

The archive distinguishes four evidential layers.

1. **Literal finite objects.** Division-free graphs and residual arrays specify the ordinary polynomials. Variable liveness, fixed-port bindings, ordered domains, operation counts and polynomial identities are checked directly.
2. **Exact algebra and data.** Independent sparse-polynomial implementations verify the seven integer transformations, centered root identities, ideal-preserving row operation, both positive norms, and actual-coefficient leading monomials. The full source34 and compact44 coefficient tuples are independently reconstructed from the 943-tile finite data rather than verified solely by a producer's checksum.
3. **Unbounded mathematical proofs.** Saturated Lucas reliability, full interpolation, rational integrality, nearest-square positivity, coefficient-height injectivity, restoration and completeness are mathematical arguments. Finite numerical fixtures do not replace them.

4. **Inherited semantic calibration.** Source/compiler/Pell interfaces connect the arithmetic sources to finite syntactic ZFC refutations. These dependencies remain explicitly retained and pinned. No newly built Lean/Rocq kernel certificate or encoded object-language ZFC derivation is claimed.

The independent ten-variable audit uses different prime moduli and nonuniform diagonal directions for exact-degree attainment. For the saturation-only control it compares the unchanged source prefix to the previous literal equation; for the recommended height-1868 equation it independently reconstructs and evaluates the changed sparse source. It verifies all actual fixed leaves, checks 480 signed modular tail assignments, 1080 local parameter cases and twelve CRT interpolation instances. The latter are deliberately finite diagnostics, not finite evidence for all indices. A separately authored checker also covers the entire source formula rather than only the tail.

The independent low-degree audit verifies all seven arrays, their exact restoration maps, explicit nonzero top coefficients, the complete four-port reconstructions and positive11/positive12 identities. The report's compact quartic, sextic and octic residual lists have a separate exact comparison against the literal packets. This catches mistakes in typeset formulas, not only in machine files.

The height certificate has its own 24,844 fixtures, 100 constructed proof trees, twelve simultaneous beta encodings and 800 residual comparisons. An independent implementation checks all 20,440 rule lists of length at most three over labels $0, 1, 2$, including polynomial equality and multiplicity recovery. The selector checks and literal Lean-constructor shape checks are additional finite evidence. Shape checks are not Lean elaboration.

6.2 Negative controls and mathematical limits

The saturated Lucas certificate is *not* a standalone index theorem without $|C| > 1$. The slice $A = 72$, $B = 31$, $C = -1$ supplies a false-index countercontrol when that hypothesis is dropped, even with the indicated residue and range conditions. In the completed consistency equation, the existing $\Delta > 0$ approximation guard proves $|C| > 1$ before reliability is invoked. This order is part of soundness, not an optional optimization.

Likewise, removing the height bound while retaining evaluated product balance is unsound: $1 + 3M = (1 + M)^2$ at $M = 1$ is a collision between different label multisets. The new register rejects it. The release tests also include mutations of coefficients and literal equations, ensuring that verification rejects known changes instead of merely replaying stored success receipts.

At the reviewed pin, ProveIt has the supplied numerical ZFC-refutation checker and an internal generic quartic compiler. The reported 83,281 raw and 76,313 optimized primitive-recursive nodes are source-program-tree counts, not equation-variable counts. Internal syntax/proof-checker transport remains open. On the native route, high-mask binding, phase synchronization and global semantic calibration have not been replaced by a complete new internal consistency biconditional. Those formal boundaries must not be inferred away from external Lean statements or finite coefficient checks.

6.3 Archive contract

The top-level `README.md` gives the precise run commands and output contract. `MANIFEST.json` lists every shipped file with its role and SHA-256. `verify.py` validates that inventory, copies the packet to an isolated temporary directory, executes only the explicitly listed authored builders and checkers, and compares generated literal artifacts byte for byte. It runs normal and optimized Python so checks must remain active under `-O`. No upstream program or refutation search is run. The original delivery copy is not modified by replay.

The TeX sources and PDF are under `report/`. Rebuild with `sh report/build.sh`; the exact mathematical replay requires Python and SymPy, while PDF rebuilding additionally requires pdfLaTeX and the listed standard packages. Native coefficient payloads are compressed minimal little-endian integers; the Lucas fixed leaves additionally carry a sign byte. These are complete integer values, not unspecified parameters. Fixed coefficients are interpreted according to each component's binding, not by filename alone.

Historical source snapshots retain their own notices, source pins and original content. Their old relative links and numerical examples describe the original packages and are not promises that every old dependency is vendored here. The new operational instructions refer only to shipped files. The release manifest identifies inert retained upstream programs; none is in the executable replay allowlist. Full arrival ZIPs, search logs, caches, abandoned drafts and copyrighted primary articles are not redistributed.

7 What to pursue next

There is no single well-defined "simplest" polynomial without a cost model. The present evidence supports the following priorities rather than a claim that the target has been minimized.

Compile the height source before optimizing its presentation. The largest architectural opportunity is the removal of an entire bounded-universal common-multiple certificate. Instantiate and elaborate the supplied constructor, prove the local relation's semantics in the chosen kernel, and measure the full bounded-forall and remainder expansion. Only then can the extra accumulator stream be weighed against the deleted block. The 23/14 local count cannot predict the global variable count by itself. This is the most direct test of whether the structural idea beats the Pell-based endpoints.

Audit how much Lucas divisibility is really needed. Norm saturation succeeds because D becomes a square in the soundness proof, while Ω still absorbs the relevant period valuations. Removing another factor from Ω or C^2 would require a new rank and full-residue interpolation theorem. The counterexample at $C = -1$ shows why a locally attractive formula cannot be accepted by gate counting alone. The correct research object is the exact divisibility interface, including exceptional indices, not an unchecked symbolic cancellation.

Reduce the source while keeping complete coefficient tuples. Compatible packing can lower degree without changing witnesses, and the factored positive gap can remove a weighted source record. These are separate levers. Any proposed replacement must preserve all extraction identities, carry bounds, positivity, canonical divisibilities and the whole fixed tuple. A lower syntactic term count without a better certified layout is not an improved final equation.

Keep degree and cost tradeoffs honest. The new sextic uses retained $F_2 = f^2$ as an ideal-preserving lifting register. Other registers may admit similar cancellations, but any elimination must retain integrality and a polynomial restoration map. For the positive equations, rational resultants need a descent or saturation argument; elimination over $\mathbb{Q}$ alone does not establish a Diophantine equivalence over $\mathbb{Z}$.

A Compact literal low degree equations

This appendix removes any ambiguity about what is meant by the completed integer packets. Each equation is $\sum_i r_i^2 = 0$ for the residuals below. The six constants are $\beta, K_0, M_C, M_F, A, B$ from `positivity/BINDING.json`, with $A = 2^{1113}$ and $B = 2^{666104}$. The symbolic A, B here are unrelated to the Lucas parameters.

A.1 The 37 integer quartic

The 37 variables are

$$J_3, F, \alpha, z, f, h, r, v, w_3, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta,$$
$$t_1, t_2, t_3, e_1, e_2, e_3.$$

Here and below line breaks in a variable list do not indicate operations. Put $J = 3J_3 + 1$, $q = \beta J + 1$, $W = 4(3w_3 + 1)$, $k = \eta + \zeta$, $H = 4a + 3$, $C = q - F - \alpha$ and

$$R = (q - F)(N - q - L) + \alpha L + (M_C + qM_F)J, \quad T = AFBC - (q - AF - BC)^2.$$

With $\mathrm{Pos}(p, q, r) = 1 + p^2 + q^2 + r(r + 1)$, the 24 residuals are

$$\begin{aligned}
&L - q^2 + 1, && N - q(L + 1),\\
&a - E - (2v + 1)N, && X - Wq,\\
&E + X(E - a), && c - \eta + (E - a)k,\\
&P - k - E(c - \eta), && \tau^2 - P^2 + Pk - 1,\\
&C_2 - c^2, && D - X - ac - \gamma H,\\
&\Delta - a^2 - H, && D^2 - \Delta C_2 - 1,\\
&k - 1 - hE - R, && (K_0 + W)C - F - z(q - 1),\\
&F_2 - f^2, && V + c - fo,\\
&V - cj + k - 1 - hE, && g + V^2 - y^2,\\
&I - (2r + 1)C_2, && B_{\mathrm{aux}} - I^2,\\
&B_{\mathrm{aux}} - \Delta(F_2 - 1), && y^2 - B_{\mathrm{aux}} g - 1,\\
&T - \mathrm{Pos}(t_1, t_2, t_3), && \eta\zeta - \mathrm{Pos}(e_1, e_2, e_3).
\end{aligned}$$

The direct 37-variable companion instead leaves J, w free, uses $X = 3X_3 - 1$, and removes X as a coordinate. Its literal rows are retained in `positivity/quartic37_direct.json`. Both versions delete the old $X = 2u^2$ guard only under the proved residue orientation.

A.2 The 30 integer sextic

Use the 30 coordinates

$$J_3, F, \alpha, z, f, h, r, v, w_3, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$E, c, a, D, L, F_2, g, I,$$
$$t_1, t_2, t_3, e_1, e_2, e_3.$$

Define J, q, W, k, H, C, T as above, $X = Wq$, and $R = ((q-1)(q-F)+\alpha)L + (M_C + qM_F)J$. The 17 residuals are

$$
\begin{aligned}
&L - q^2 + 1,\\
&a - E - (2v+1)q(L+1),\\
&E + X(E-a),\\
&c - \eta + (E-a)k,\\
&\tau^2 + (2\tau - k)E(c-\eta) - 1,\\
&D - X - \gamma H,\\
&D^2 + 2acD - Hc^2 - 1,\\
&k - 1 - hE - R,\\
&(K_0 + W)C - F - z(q-1),\\
&F_2 - f^2,\\
&fo - c - cj + k - 1 - hE,\\
&g + F_2 o^2 - 2foc + c^2 - y^2,\\
&I - (2r+1)c^2,\\
&I^2 - (a^2 + 4a + 3)(F_2 - 1),\\
&y^2 - I^2 g - 1,\\
&T - \mathrm{Pos}(t_1, t_2, t_3),\\
&\eta\zeta - \mathrm{Pos}(e_1, e_2, e_3).
\end{aligned}
$$

The coordinate called D here is the centered root, not the old D. Its restoration is $D_{\mathrm{old}} = D + ac$; likewise $\tau_{\mathrm{old}} = \tau + E(c-\eta)$.

A.3 The 26 integer octic and its three companions

Use 26 variables

$$
\begin{aligned}
&J_3, F, \alpha, z, f, h, r, v, w_3, \tau, \eta, y, \gamma, o, j, E, c, a, V, I,\\
&t_1, t_2, t_3, e_1, e_2, e_3.
\end{aligned}
$$

Put

$$
\begin{gathered}
J = 3J_3 + 1, \quad q = \beta J + 1, \quad W = 4(3w_3 + 1), \quad X = Wq,\\
C = q - F - \alpha, \quad k = 1 - V + jc + hE, \quad H = 4a + 3, \quad \Delta = a^2 + H,\\
D = X + ac + \gamma H, \quad P = k + E(c - \eta),\\
R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
T = AFBC - (q - AF - BC)^2, \qquad Z_+ = \eta(k - \eta).
\end{gathered}
$$

The 13 residuals are

$$
\begin{aligned}
&a - E - (2v+1)q^3, && E + X(E-a),\\
&c - \eta + (E-a)k, && \tau^2 - PE(c-\eta) - 1,\\
&D^2 - \Delta c^2 - 1, && k - 1 - hE - R,\\
&(K_0 + W)C - F - z(q-1), && V + c - fo,\\
&I - (2r+1)c^2, && I^2 - \Delta(f^2 - 1),
\end{aligned}
$$

$$I^2(V^2-y^2)+y^2-1, \qquad T-\mathrm{Pos}(t_1,t_2,t_3),$$
$$Z_+-\mathrm{Pos}(e_1,e_2,e_3).$$

The following literal substitutions define the companions without any implicit existential register:

1. For 25 variables and degree 10, replace α by the free coordinate C, substitute $F=(K_0+W)C-z(q-1)$ and $\alpha=q-F-C$ everywhere, and delete the transport residual and the coordinate F.
2. For 24 variables and degree 12, delete the six sign coordinates and the two sign residuals, insert four coordinates s,p,q_+,r_+, and insert the one residual
$$TZ_+-(s^2-T-Z_++1)^2-\mathrm{Pos}(p,q_+,r_+).$$
3. For 23 variables and degree 16, perform both operations in that order or the reverse order. Their existence proofs and exact degree certificates were given in the main text.

The JSON packets use `v_scale`, `cone_A`, `cone_B`, and `t` where this appendix uses v,A,B,s. These are renamings only. Their ordered variable lists are authoritative for evaluators.

B The exact sparse source kernels

These are the fixed source polynomials used in Section 2. The four coefficients β,K_0,M_C,M_F here are the whole original31 tuple in `lucas_h1868/data/coefficients_binding.json`, not the compact44 tuple of the positive equations. Their actual unsigned integer bytes are in the adjacent compressed payloads. Derived signed fixed leaves are in `lucas_h1868/fixed/`; `FIXED_COEFFICIENTS.json` specifies the sign-byte convention, exact raw and compressed hashes, and bit lengths. The literal graph also retains the exact integer-polynomial recipes for every derived leaf. No undetermined parameter remains.

Weights and target exponents. The 31 weights, in source variable order, are

$$1,7,61,36,16,85,20,67,43,11,34,13,96,80,29,50$$
$$2,109,110,112,116,91,53,55,27,101,92,9,107,37,60$$

The 22 target exponents, in residual order, are

$$1141,28,1437,1069,1836,1354,1814,614,267,1408,1403$$
$$889,337,1587,413,673,1139,772,1131,1018,1055,85$$

For each $d=0,1,2$, define $D_d(B)=\sum_e c_{d,e}B^e$ using the following complete lists. A missing exponent has coefficient zero.

Layer d	Exponent e	Coefficient $c_{d,e}$
0	85	-2
0	337	-2
0	413	$2K_0$
0	614	-2
0	889	-2
0	1408	-6
1	1	-2
1	26	2

Layer d	Exponent e	Coefficient $c_{d,e}$
1	27	-2β
1	176	-2
1	214	2
1	217	-6
1	241	2
1	245	2
1	257	2
1	322	2
1	352	$-2K_0$
1	406	$-2K_0 - 2$
1	412	$2K_0\beta + 2\beta$
1	572	2
1	763	2
1	953	2
1	960	-2
1	981	2
1	995	2
1	1018	2
1	1024	2
1	1027	2
1	1047	2
1	1058	-2
1	1114	2
1	1140	-4β
1	1242	2
1	1258	-2
1	1292	-8
1	1346	2
1	1348	2
1	1403	-2
1	1495	-2
1	1586	$-2M_C - 2M_F$
1	1704	2
1	1727	2
2	0	$-\beta$
2	27	2
2	39	-1
2	101	-4
2	143	-1
2	182	-1
2	318	-1
2	372	-1
2	376	$-\beta$
2	394	-2
2	408	-1
2	424	-1
2	588	2

Layer d	Exponent e	Coefficient $c_{d,e}$
2	641	-2
2	714	-2
2	774	-1
2	783	2
2	804	-2
2	894	-1
2	1056	-1
2	1139	$-2\beta^2$
2	1142	-1
2	1149	1
2	1158	-1
2	1165	1
2	1176	-2
2	1179	-2
2	1402	$-\beta$
2	1432	1
2	1499	-1
2	1553	-1
2	1578	1
2	1579	$-\beta$
2	1584	$-\beta$
2	1585	$-2M_F\beta + 2\beta^2$
2	1593	-1
2	1609	1
2	1629	-1
2	1636	1

The certified layout has 86 coefficient records sharing 81 stored tap positions, $W = 116$, height 1868 and maximum target 1836. Its layer degrees are 1408, 1727 and 1636, with leading coefficients $-6, 2, 1$. The dominance gaps are 460,25,0 and target headroom is 32. Every coefficient value is retained, so the original bound $2^{2\,134\,927}$ remains valid; the fixed extraction radix satisfies $B_0 \geq 4096 \cdot 2^{2\,134\,927} b^2$ whenever the source scale is positive. The separate factored-gap source in `packing/factor_gap/` reduces 86 weighted records to 85 by an exact positive zero-set bijection. No better completed packing endpoint is claimed for that source, and it is not silently substituted into the recommended equation.

C Sources and attribution

[1] ProveIt, reviewed main commit `2605669a532342f153b13eb6b4a0af4b54f5fa37`. `https://github.com/VladimirReshetnikov/ProveIt/tree/2605669a532342f153b13eb6b4a0af4b54f5fa37/Computability/DiophantineConsistency`. The intake, placement and review commits are recorded in `review/WAVE10_REVIEW.md` and the associated pin manifests.

[2] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Theorems 4.12–4.13. `https://arxiv.org/html/1704.03504v7`.

[3] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, modified Lucas certificate and interpolation; Lemmas 2.1–2.2, Corollary 2.1, Theorems 3.1 and 4.1, and Propositions 6.1–6.2. `https://arxiv.org/html/2609.01594v1`.

[4] J. P. Jones, D. Sato, H. Wada and D. Wiens, *Diophantine representation of the set of prime numbers*, American Mathematical Monthly 83 (1976), 449–464. `https://www.cs.umd.edu/~gasarch/BLOGPAPERS/Jonesh10.pdf`. The article is cited, not redistributed.

[5] The retained source68 free-Lucas and commutative-flow proofs, source69 rational-Pell proof, source70 weighted transport and three-root proof, source71 calibrated determinant guard, source72 sextic lifting choice, and source73 adaptive carrier are the direct compositional interfaces. Their exact source members are listed in `DEPENDENCIES.md` and `MANIFEST.json`.

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